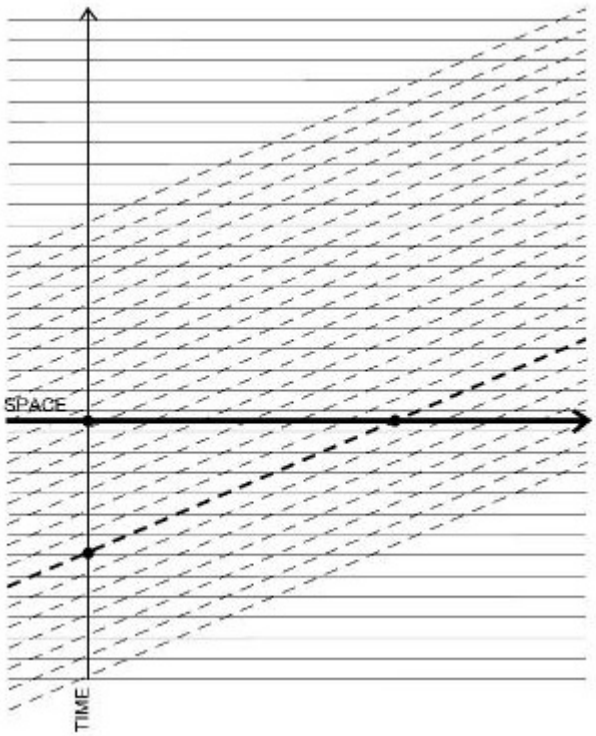


The light solid lines are equal time slices (now-slices) for observers at rest with respect to earth (for simplicity, we imagine that earth is not rotating or undergoing any acceleration, as these are irrelevant complications for the point being made), and the light dotted lines are equal time slices for observers moving away from earth at, say, 9.3 miles per hour. When Chewie is at rest relative to earth, the former represent his now-slices (and since you are at rest on earth throughout the story, these light solid lines always represent your now-slices), and the darkest solid line shows the now-slice containing you (the left dark dot), in earth's twenty-first century, and he (the right dark dot), both sitting still and reading. When Chewie is walking away from earth, the dotted lines represent his now-slices, and the darkest dotted line shows the now-slice containing Chewie (having just gotten up and started to walk) and John Wilkes Booth (the lower left dark dot). Note, too, that one of the subsequent dotted time slices will contain Chewie walking (if he is still around!) and you, in earth's twenty-first century, sitting still reading. Hence, a single moment for you will appear on two of Chewie's now-lists—one list of relevance before and one of relevance after he started to walk. This shows yet another way in which the simple intuitive notion of *now* —when envisioned as applying throughout space—is transformed by special relativity into a concept with highly unusual features. Furthermore, these now-lists do *not* encode causality: standard causality (note 11, Chapter 3) remains in full force. Chewie's now-lists jump because he jumps from one reference frame to another. But every observer—using a single, well-defined choice of spacetime coordinatization—will agree with every other regarding which events can affect which.

4. The expert reader will recognize that I am assuming spacetime is Minkowskian. A similar argument in other geometries will not necessarily yield the entire spacetime.

5. Albert Einstein and Michele Besso: Correspondence 1903–1955, P. Speziali, ed. (Paris: Hermann, 1972).